

Antecedents to responsible financial management behavior among young adults: moderating role of financial risk tolerance

The role of
financial risk
tolerance

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Abstract

Purpose – The study examines the antecedents of responsible financial management behavior among young adults in India and explores the role of financial risk tolerance as a moderating variable.

Design/methodology/approach – The sample includes young adults in the age group of 18–35. The analysis uses a two-step approach via standard partial least squares structural modeling (PLS-SEM) and ordinary least square (OLS) regression.

Findings – Structural modeling results show that financial attitude fully mediates the relationship between financial knowledge and responsible financial management behavior, and locus of control influences responsible financial management behavior. Financial risk tolerance moderates the relationship. Among demographic factors, age and occupation influence responsible financial management behavior.

Research limitations/implications – The financial knowledge used in the survey are based on self-reported responses. The future study can include participants from both developed and emerging countries to assess similarities and differences.

Practical implications – Despite the growing focus on improving financial literacy, there are growing concerns regarding responsible financial behavior. Since financial services is related to fiduciary responsibility, managers and policymakers need to ensure that financial knowledge results in improving financial attitude, which further leads to responsible financial behavior.

Originality/value – The present study from an emerging country will add value to the literature.

Keywords Responsible financial management behavior, Financial knowledge, Financial attitude, Internal locus of control, Financial risk tolerance

Paper type Research paper

Introduction

Policymakers and financial services marketers have focused on measures to improve financial decision making and financial management behavior across the nations. A variety of factors play an important role in responsible financial management behavior, including global downturn, the need to maintain the savings rate which has the likelihood of moving southwards and a shift toward credit culture. While loan defaults, particularly in developed countries, caught attention after global financial crisis, policymakers and regulators need to address the concern for an emerging country like India, where the banking industry is facing asset quality issues among both priority and non-priority segment (RBI Annual Report). The shift in global finance away from US and European countries and growing interest in emerging countries is also a reason to examine financial management behavior.

The study contributes to the literature on financial management behavior in various ways. First, while most of the past studies have investigated financial management behavior among young adults from a developed country perspective, few studies delve into the topic from an emerging country perspective. The present study fills the gap by coverage on respondents from India. Second, experience has also raised questions regarding the efficacy of financial literacy (Alsemgeest, 2015; [Reyers, 2019](#)). It is observed that contrasting results are available regarding the role of financial literacy efforts toward responsible financial



management behavior, particularly among young adults. Third, consistent with the financial management model covering input, personal subsystem and managerial subsystem which is an adaptation of [Deacon and Firebaugh \(1988\)](#) model, the present study considers that financial knowledge will influence financial attitude which will further impact financial management behavior. In addition to financial knowledge and financial attitude, the study examines the role of locus of control on responsible financial management behavior. Fourth, most of the earlier studies have relied on the financial management scale by [Fitzsimmons *et al.* \(1993\)](#) and [Prochaska-Cue \(1993\)](#). Thereafter, the scale developed by [Dew and Xiao \(2011\)](#) was psychometrically validated. The scale is comprehensive and generalized since it covers the aspects of savings, investment, cash management and credit management. The study has employed the scale developed by [Dew and Xiao \(2011\)](#). And fifth, past studies have explored financial risk tolerance, with demographic, socioeconomic and attitudinal characteristics as independent variables ([Grable, 2000](#)). The purpose of the study is to extend the inquiry by classifying financial risk tolerance into higher and lower financial risk-tolerant. In addition to covering major antecedents such as financial knowledge, financial attitude and internal locus of control, the study examines financial risk tolerance as a moderator to financial management behavior. The objective of the study is to explore antecedents to financial management behavior among young segments in an emerging country.

The next section covers the theory and conceptual framework. Thereafter, hypothesis development is discussed. It is followed by data collection, methodology, results, discussion, limitations, conclusions and managerial implications.

Theory and conceptual framework

The topic is important to financial services since financial services are related to fiduciary responsibility. Fiduciary responsibility requires that financial services marketers discharge duties with a focus on the client's interest. Since depositors repose trust and faith by keeping money with banks/financial institutions, financial services marketers have a responsibility to protect depositor's interest. Responsible financial management behavior, which deals with savings and investments, and changing asset allocations when appropriate, can play an important role in fiduciary responsibility. With growing concerns of low financial literacy ([Agarwalla *et al.*, 2015](#)), increased sophistication of financial products and issues associated with incidences of misselling, there is greater attention to financial management behavior.

According to the transtheoretical model of change, both knowledge and psychological factors contribute to behavioral change ([DiClemente and Prochaska, 1982](#)). The importance of psychological aspects is recognized in the theory of planned behavior ([Ajzen, 1991](#)), which examined the attitude-behavior relationship. According to the theory of planned behavior, among other factors, attitude shapes behavior intentions and behaviors. [Shim *et al.* \(2010\)](#) observed a positive relationship between attitude and financial management behavior. The transtheoretical model of behavior change holds that self-efficacy, which is related to internal locus of control, plays a crucial role in behavioral change ([Prochaska and DiClemente, 2005](#)). The uncertainty theory relates to individual risk aversion to the wealth that an individual possesses. Using the theory of uncertainty, [Mas-Colell *et al.* \(1995\)](#) and [Varian \(1992\)](#) pointed out that individuals tend to exhibit lower risk and start taking a risk once their income crosses the threshold level.

The financial situation in young adults is characterized by a high level of debts, which warrants greater attention. The increasing debt burden requires better management of money resources to navigate financial decisions ([Lusardi *et al.*, 2010](#)). Young adults, including students and those who join the working class, confront complicated financial decisions in today's demanding environment. With the given situation where public sector institutions are reducing pension burden, it becomes more challenging. In emerging countries, there is increased availability of debt through higher exposure from both traditional and digital finance and

development of newer methods of financing, like P2P financing through electronic means. Given that young adults comprise the majority population and there are increased concerns regarding responsible financial management behavior, the growing importance of young generation necessitates insights into antecedents to responsible financial management behavior.

Financial knowledge

An individual acquires knowledge through varied sources such as emergencies, observations and conceptualizations (Tang and Peter, 2015). Varied conceptual definitions about financial knowledge are available, ranging from training in financial management (Godwin, 1994, p 172), to completion of a consumer education course (Godwin and Carroll, 1986, p 81) to multi-item index of knowledge (Hira *et al.* 1992; Titus *et al.*, 1989). Financial knowledge is referred to as the possession and comprehension of information about financial matters. Financial knowledge plays a critical role in comprehending essential financial concepts and products (Chen and Volpe, 1998). Financial literacy, financial knowledge, financial education and financial capability are used interchangeably in the literature (Atkinson *et al.*, 2006; Xiao *et al.*, 2014). Alba and Hutchinson (2000) suggested that financial knowledge is divided into two components: subjective knowledge, which refers to consumer assessment and objective assessment, which relates to accurate information. Subjective knowledge is about belief regarding what he/she knows and objective knowledge relates to what an individual knows (Tang and Baker, 2016; Nejad and Javid, 2018). Financial knowledge was identified as the major determinant of wealth inequality (Lusardi *et al.*, 2017).

Past studies have raised concerns regarding low financial knowledge (Cude *et al.*, 2019; Kim *et al.*, 2019). Although financial knowledge can be imparted at different stages of life, Lusardi *et al.* (2013) stressed the importance of imparting financial knowledge during an early stage in the working career. Consumers who lack adequate financial knowledge exhibit greater vulnerability. To overcome the low financial knowledge, newer methods like video intervention was found to be successful (Kuntze *et al.*, 2019). The antecedents to financial knowledge include individual preferences, household resources and the cost of acquiring financial knowledge. The studies investigating the role of gender are mixed. While Fletschner and Mesbah (2011) indicated that men possess more financial knowledge than women, Swamy (2014) noted that families, where women participate in financial decisions, have a greater percentage rise in family income.

Financial attitude

Attitude is the evaluation of ideas, event, objects or people and plays an important role in predicting consumer behavior in various settings. Financial attitude is considered as the preference and dispositions toward personal finance issues (Aydin and Sycuk, 2019) and is referred to as the psychological tendency expressed by evaluating financial management with some level of agreement or disagreement (Parrotta and Johnson, 1998). Few studies have examined financial attitudes (Godwin, 1994; Godwin and Koonce, 1992; Wilhelm *et al.*, 1993), and confirmed the importance of attitude in determining means of money. Financial attitude has been treated as an independent variable (Godwin, 1994) and dependent variable (Godwin and Carroll 1986; Godwin and Koone, 1992). Past studies indicate that financial attitude is shaped between the ages of 17 and 21 (Meredith and Schewe, 1994). Financial attitude varies based on the region (Jorgensen *et al.*, 2017) and culture (Tang, 1993).

Internal locus of control

Locus of control refers to person's perception of life and relates to the degree to which success or failure is a result of their action (Chiu, 2003; O'Connor and Kabadayi, 2019) Rotter (1966)

introduced the topic of locus of control, which is a measure of an individual's belief about cause and effect within the life. Locus of control is divided into the external and internal locus of control. While the external locus of control deals with fate, luck and chance, internal locus of control is focused on skill, outcome and knowledge which results in the desired outcome. The seminal work by [Perry and Morris \(2005\)](#) considered locus of control as an antecedent to financial management behavior. The consideration of locus of control allows for the inclusion of cultural and psychosocial perspective. Locus of control was found to be different from varying ethnic background as some culture depicts varying risk tolerance levels. According to [Grable et al. \(2009\)](#), since the internal locus of control tends to be goal-driven, it results in responsible financial behavior.

Financial risk tolerance

Risk can be categorized as a monetary, physical, psychological and social risk. Monetary risk deals with potential financial loss; social risk is about self-esteem, reputation and perception of others; physical risk relates to injury to the body and psychological risk happens when a customer exhibits higher potential reward for investment despite the possibility for a loss. [Havlena and Desarbo \(1991\)](#) considered perceived risk emanating from uncertainty of possibility from a negative outcome. Financial risk tolerance is defined as the maximum amount of uncertainty that someone is willing to accept when making a financial decision ([Grable, 2000](#)). [Slovic \(1986\)](#) considered that risk assessment is subjective and influenced by various variables including psychological, social, cultural and political factors. Various demographic factors affect an individual's risk tolerance ([Hallahan et al., 2004](#)). [Wallach and Koyan \(1961\)](#) found that risk levels differ with age, with younger individuals ready to take a higher level of risk. [Wang and Sherman \(1997\)](#) explored the role of life cycle data using sources from consumer finances. The study by [Hsee and Weber \(1999\)](#) confirmed the impact of cultural background on risk tolerance while comparing Chinese and American culture. Past studies have also explored the role of risk tolerance in investment decisions ([Yip, 2000](#)).

Responsible financial management behavior

The actual level of consumer debt has been considered a proxy for measuring financial management behavior during the early periods. [Hilgert et al. \(2003\)](#) proposed that individuals follow a hierarchical pattern. Variations are observed in individuals following cash flow, savings, credit and investment. The reasons for variations can be attributed to differing financial resources and competencies. Financial management behavior covers the domain of cash, savings, credit and investment management. Each of the domains has certain consequences, and it is important to include all the domains. Sound financial management behavior has personal and non-personal consequences, such as improved physical health, better mental health and life satisfaction ([Xiao et al., 2011](#)). The need for comprehensive and psychometrically strong measures of financial management behavior was felt by researchers and practitioners. [Grable et al. \(2009\)](#) found cross-cultural differences in the behavior.

Hypothesis development

Financial knowledge and financial attitude

Most of the studies on young adults have found low financial knowledge among young adults. The financial knowledge was found to be much below the expectations of policymakers, financial counselors and government bodies. While students are lured toward credit cards due to aggressive marketing, they do not understand the implications of acquiring debt. As a result, concerns are raised regarding money management skills ([Norvilitis et al., 2006](#)). [Warwick and Mansfield \(2000\)](#) found that that half of students

reported issues with a credit balance and credit limit. It is likely that poor financial knowledge will lead to inadequate money management skills, resulting in a low financial attitude. Individuals who are equipped with a higher level of financial knowledge have a tendency to develop a favorable attitude (Lim *et al.*, 2014; Aydin and Selcuk, 2019). There is a significance of financial knowledge in the financial well-being provided that financial knowledge is translated into favorable attitude of decision makers. We posit the hypothesis as

H1. Financial knowledge positively influences financial attitude.

Financial attitude and responsible financial management behavior

Tsui-Yii and Sheng-Chen (2014) identified the antecedents to financial attitude as power prestige, retention, anxiety achievement and respect. Funfgeld and Wang (2009) identified five underlying dimensions of financial attitudes and behavior as anxiety, interests in financial issues, decision styles, need for precautionary saving and spending tendency. Most of the earlier studies have explored antecedents to financial attitude, but few studies could explore consequences to financial attitude. Gudmunson and Danes (2011) found that financial socialization is an antecedent to financial attitudes, which in turn influences financial behaviors. Similarly, poor financial attitude can result in less desirable financial behavior. Parrota and Johnson (1998) found that financial attitude, but not financial knowledge, results in desired financial management practices. It further confirmed that financial knowledge does not mediate the relationship between financial attitude and financial practices. In recent times, the linkage between attitude and behavioral change for other products is observed and this trend is also applicable to financial situations (Jorgensen *et al.*, 2017). The financial attitude – behavior link is also consistent with the theory of planned behavior.

H2. Financial attitude positively affects responsible financial management behavior

Financial knowledge and responsible financial management behavior

Researchers often emphasize the role and importance of financial knowledge. Considering its influence on financial management behavior, policymakers and educators have introduced it as a formal curriculum in school and college education. Park *et al.* (1988) emphasize that perceived knowledge plays an important role in cognitive functioning, including recognition (Schachter, 1983), identification (Nelson *et al.*, 1984) and problem-solving (Metcalf, 1986). An individual's actual financial knowledge contribute to investments (Kyrychenko and Shumb, 2009), retirement planning (Parker *et al.*, 2012) and credit card behaviors (Allgood and Walstad, 2013). Carpenna *et al.* (2011) suggested that financial knowledge contributes in building awareness and familiarity with financial services and products. Although contrasting results are found while examining the relationship between financial knowledge and behavior with some studies showing positive relationships (Chen and Volpe, 1998; Lusardi and Mitchell, 2007), others did not exhibit the relationship (Borden *et al.*, 2008; Jones *et al.*, 2005). Tokar (2015) supported the role of financial knowledge along with confidence in shaping financial management behavior. Based on the theoretical rationale and some empirical evidence, we suggest the hypothesis:

H3. Financial knowledge positively influences responsible financial management behavior

Internal locus of control and responsible financial management behavior

Rottler (1966) introduced the concept of the locus, which can be distinguished majorly into an external and internal locus of control. Various factors such as fate, luck and choice affect external locus of control, and individuals who exhibit higher external locus of control rely on outside forces and actions. Internal locus of control is associated with skill, ability and knowledge. Perry and Morris (2005) examined consumers' willingness to budget and control

spending using a dual psychological and cultural perspective and found a negative relationship between external locus of control and financial management behavior. [Perry \(2008\)](#) observed that consumers with an internal local of control have higher credit scores. Internal locus of control showed a positive relationship with academic achievement and psychological well-being ([Lefcourt, 1991](#)) and job performance ([Judge and Bono, 2001](#); [Spector, 1982](#)). Internal locus of control reported a positive influence on financial preferences and behavior ([Hira and Mugenda, 2000](#); [Onkvisit and Shaw, 1987](#); [Prince, 1993](#)). We posit the hypothesis that

H4. Internal locus of control has a positive influence on responsible financial management behavior.

Financial risk tolerance as a moderating variable

The concept has been studied from the perspective of risk aversion ([Arrow, 1971](#)) and risk tolerance ([Barsky et al., 1997](#); [Gron and Winton, 2001](#); [Grable, 2000](#)). [Pratt \(1964\)](#) suggested that people who were more risk-averse would invest a smaller portion of wealth in a risky asset. [Grable \(2000\)](#) explored how age, education, income, amount of financial knowledge and economic expectations had positive effects on respondents' risk tolerance. Financial risk is important in financial services industry. Since the perceived risk and level of uncertainty are high in financial services industry ([Carlsson and Nilsson, 2017](#)), it is relevant to study financial risk and relate it with financial management behavior. Risk has been considered a moderator in the context of satisfaction with loyalty, product evaluation with choice ([Gurhan-Canli and Batra, 2004](#)) and incongruity with evaluation ([Campbell and Goldstein, 2001](#)). [Worthy et al. \(2010\)](#) confirmed the relationship between financial risk and financial management behavior in the context of credit cards where there are issues of financial debts. We argue that financial risk tolerance can be a moderator toward financial management behavior. We propose the following hypothesis:

H5. The relation between financial knowledge, financial attitude and internal and responsible financial management behavior is moderated by financial risk tolerance.

We present the model as follows ([Figure 1](#)):

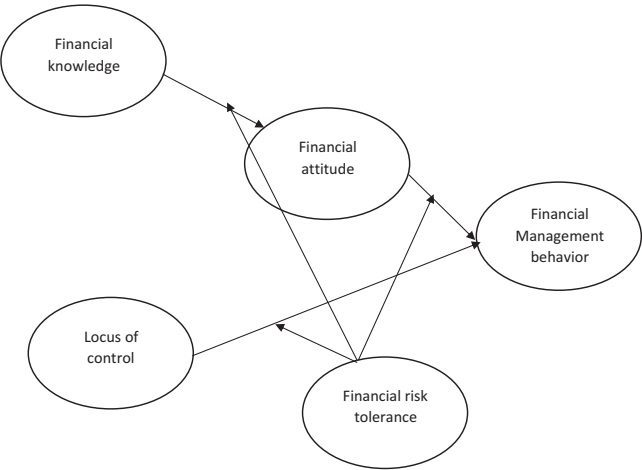


Figure 1.
Antecedents to
financial management
behavior

Methodology

Data collection and sampling

Our study is focused on young adults who are in the age group of 18–35. Since there is a sizeable proportion of young adults who use the Internet, it was decided to collect data through email or obtain a response on the link. Cross sectional survey data using a structured questionnaire were used. Since the respondents followed English as a medium of instruction, the questionnaire was subject experts who were drawn from the financial services domain. The respondents included students and working professionals. Convenient sampling was followed to collect the responses. In total, 584 responses were received with the following characteristics: 60% male and 40% female with average age of 25 years. The sample covered 48% who were employed and 52% who were working as students. About 35% of respondents have studied finance as a formal subject.

Measures

In the past, differing scales have been used for financial management behavior. As the scale of financial management behavior has evolved, it was decided to include the scale used by [Dew and Xiao \(2011\)](#). We found that the scale was comprehensive, covering the comprehensive aspects of cash management, credit management and savings and investment. Financial attitude is referred to as the personal disposition toward personal financial matters. The scale, consisting of six items, was drawn from the scale developed by [Chen and Volpe \(1998\)](#). The scale on internal locus of control covered was adapted from that developed by [Grable et al. \(2009\)](#). The scale on financial knowledge was adapted from [Perry and Morris \(2005\)](#). The scale on financial risk tolerance was adapted from the scale developed by [Nga et al. \(2010\)](#). After finding the scores, we divided the financial risk tolerance into two levels, with an average score of more than and equal to 4.5 was classified as financially intolerant and other respondents were considered as financial tolerant. All the above scales were measured on a Likert scale with 1 being strongly disagree to 5 being strongly agree. [Table 1](#) shows constructs, related items and calculated scores for Cronbach's alpha, composite reliability, average variance extracted (AVE) and factor loadings of each item.

Analysis and results

Evaluation of measurement scales

To analyze the proposed model, a two-step approach via standard partial least squares structural modeling (PLS-SEM) using Smart PLS was followed ([Ringle et al., 2015](#)). The study first evaluated the reflective measurement models for the reliability and validity of the sample. The application of PLS is relevant at a time when the objective is to advance theoretical arguments and the study is intended to predict ([Hair et al., 2012](#)).

Reliability and validity

The evaluation of measures followed the broad guidelines prescribed by [Hair et al. \(2006\)](#). Cronbach's alpha is a popular measure of reliability. It is required that the value of Cronbach's alpha is more than 0.7, meeting the requirements of reliability.

Validity is referred to as the degree to which it accurately measures the characteristics it is made to measure and consist of content validity and construct validity. Content validity is known as face validity and is an evaluation of how adequately the measure assesses the domain of the construct ([Churchill, 1979](#)). To confirm content validity, it is required for the researcher to comprehend the domain of the measure for which exhaustive literature is carried out. The positive confirmation from experts, who were both academicians and

Table 1.
Measurement model
results

Construct	Measurement items	Factor loading
Financial Management behavior (composite reliability = 0.894, Cronbach's alpha = 0.882, AVE = 0.504)	I compare prices when I purchase a product or service 1.	0.701
	I pay my bills on time 2.	0.702
	I keep a record of my monthly expenses 3.	0.729
	I stay within budget or spending plan 4.	0.731
	I pay my credit card balance in full each month 5.	0.707
	I compare prices when I purchase a product or service 6.	0.705
	I return borrowings on time if I borrow from friends 7.	0.702
	I maintain an emergency savings fund 8.	0.765
	I save money from every paycheck or allowance 9.	0.779
	I save for a long-term goal, such as a mobile, car, education, home, etc.	0.728
Financial knowledge (composite reliability = 0.882, Cronbach's alpha = 0.859, AVE = 0.600)	I know about interest rates charged by banks or financial institutions.	0.704
	I know about credit score or ratings done by banks companies and why it is done.	0.706
	I know about basics of managing personal finance.	0.777
	I know about details of investment.	0.797
	I clearly understand the details of my bank statement.	0.874
Financial attitude (composite reliability = 0.882, Cronbach's alpha = 0.838, AVE = 0.558)	It is important for me to develop a regular pattern of saving.	0.767
	Keeping records of financial matters is beneficial.	0.782
	Financial planning for retirement is necessary for one's security during old age.	0.847
	It is beneficial to insure against reasonable risks.	0.804
	I have clear financial goals that help me determine priorities in spending.	0.703
	I believe that financial planning for 5 or 10 years in the future is essential for financial success.	0.704
	I can solve many of my problems.	0.870
Internal locus of control (composite reliability = 0.847, Cronbach's alpha = 0.761, AVE = 0.650)	I can change the important things in my life by myself.	0.761
	I am the master of my destiny.	0.784

practitioners, validated face validity. In our study, these experts were academicians from business schools dealing with banking and behavioral finance subjects. Practitioners consisted of executives who dealt with customers in the area of financial counseling.

Common method bias is problematic in quantitative studies and in studies involving self-reporting. Conformity to test for common method bias is required to meet the conditions of validity. Procedural design and statistical control are two requirements to reduce the occurrence of common method bias. Based on some of the recommendations from Podsakoff *et al.* (2003), reasonable care was taken while designing a questionnaire. For a statistical test, Harman's one-factor test was conducted. Since the single factor extraction was 30.248, which was less than the threshold value of 50 percent, the results confirm that there is no threat of common method bias.

Construct validity consists of convergent and discriminant validity. For meeting convergent validity, it is required that items positively correlate with other items of the same construct (Churchill, 1979). Since all the values of the AVE exceeded the value of 0.50,

convergent validity was established. To assess the discriminant validity, we verified for cross-loading using Fornell-Larcker criteria. For our study, we found that all the latent loadings were higher than the cross-loadings of other latent variables. Furthermore, it was observed that the AVE exceeded the corresponding squared correlations among the latent constructs (Table 2)

Structural model results

Partial least squares approach to structural equation modeling (PLS-SEM) is a variance-based approach to examine the relationship among the variables simultaneously (Chin, 1998). All these constructs together explained 38.40 percent by referring to the value of R square. To understand the mediating effect of financial attitude on the relationship between financial knowledge with financial management behavior, the study followed the simultaneous assessment of mediation effect as suggested by Iacobucci and Duhachek (2003). There is a requirement of meeting the conditions of the mediating test, which include that predictors have a significant influence on the mediator as well as the criterion variable, and the mediator has a significant influence on the criterion variable. The study meets the criteria to perform the mediation analysis (Refer Table 3). The path coefficient of financial knowledge, financial attitude, and locus of control was significant and positive. Sobel's (1982) Z-test was used to verify the mediation effect of the research model. If the Z-value exceeds 1.96 at the 5% significance level, there is an indirect effect. The path coefficient between financial knowledge and financial attitude was significant and that between financial attitude and financial management behavior was significant. However, the path coefficient of financial knowledge and financial management behavior is not significant. Thus, we can claim that the mediator is consistent with the hypothesized theoretical framework, confirming that financial attitude fully mediates the relationship between financial knowledge and financial management behavior (Table 4). The moderating impact of risk tolerance was assessed through multi-group analysis using an average of two statements: "I am in favor of making investments in shares" and "For me, returns is more important than safety." Higher scores indicated more propensity for risk and hence classified as financial intolerant, and lower score was refereed as financial tolerant. The magnitude of path coefficients for high risk was higher than that of low risk for the path financial knowledge- financial and the magnitude of path coefficient for low risk was higher than high risk for the path financial attitude- responsible financial management behavior and internal locus of control-responsible financial management behavior. Thus, findings suggest that risk tolerance acts as a moderator for both the types of risk but only with different levels of magnitude (Table 5).

	Financial management behavior	Financial attitude	Financial knowledge	Internal locus of control
Financial management behavior	0.697			
Financial attitude	0.122	0.782		
Financial knowledge	0.596	0.348	0.748	
Internal locus of control	0.381	0.249	0.475	0.806

Note(s): Diagonal item shows average variance extracted (AVE) and other values represent squared interconstruct correlations

Table 2.
Discriminant validity
results

Regression results

We performed ordinary least square (OLS) regression to analyze the impact of demographic variables such as gender, age, academic background and occupation on dependent variables, responsible financial management behavior. As observed in Table 6, we find that age and occupation significantly affect responsible financial management behavior. While further conducting analysis of variance (ANOVA) to assess the difference in mean, we find that means of higher age = 4.20; medium age = 4.04 and low age = 3.69; and means of working people = 4.05 and student = 3.67 were significantly different. Although the value of score for responsible financial management behavior for male (3.90) was higher than that for female (3.70), they were not statistically different. Similarly, the mean score of individuals with finance background (3.90) was not statistically different from that without a finance background (3.77).

Discussion

Our study contributes to the existing literature on financial management behavior in multiple ways. First, the study applies the representative, comprehensive, psychometrically validated scale developed by Dew and Xiao (2011), which covers the aspect of savings, investment and credit management. Second, the study identifies the antecedents to financial management behavior and shows how financial knowledge, financial attitude and internal locus of control influence responsible financial management behavior. The present study confirms that financial attitude is a mediator between financial knowledge and responsible financial

Table 3.
Structural model
results without
mediator

Constructs	Coefficients	Standard error	T Statics	Result
Financial knowledge -> Financial attitude	0.010	0.007	2.701	Significant
Financial attitude -> financial management behavior	0.578	0.000	14.470	Significant
Internal locus of control -> Financial management behavior	0.133	0.001	3.377	Significant

Table 4.
Structural model
results with mediator

Constructs	Coefficients	p value	T Statics	Result
Financial knowledge -> Financial attitude	0.372	0.000	3.878	Significant
Financial attitude -> financial management behavior	0.533	0.000	6.581	Significant
Financial Knowledge -> financial management behavior	-0.084	0.117	1.02	Not Significant
Internal locus of control -> Financial management behavior	0.228	0.013	2.489	Significant

Table 5.
Multi-group analysis
result: financial risk
tolerance

Constructs	Global model	High risk	Low risk	Supported
Financial knowledge -> Financial attitude	0.372	0.380	0.297	Yes
Financial attitude -> financial management behavior	0.533	0.513	0.643	Yes
Internal locus of control -> Financial management behavior	0.228	0.141	0.325	Yes

management behavior. Financial knowledge does not directly influence financial management behavior. The findings are consistent with the recent study by Tang and Peter (2015), Borden *et al.* (2008) and Jones *et al.* (2005), which confirmed weak or no relationships between financial knowledge and financial behavior. This suggests that the individual may not take full advantage of their financial knowledge unless it results into developing appropriate financial attitude. Internal locus of control is considered a variable that affects responsible financial management behavior. As the magnitude of the path coefficient of financial attitude was higher than the internal locus of control, financial attitude deserves more importance. The present study differed from Grable *et al.* (2009) as the study did not establish a direct relationship between financial knowledge and financial management behavior.

Limitations

We would also like to highlight the limitation of this study. The financial knowledge used in the survey is based on self-reported responses. These questions may fail to capture respondents' financial knowledge in other financial areas or their application-oriented knowledge. The future study can add the construct pertaining to financial literacy. As measures of financial literacy become more rigorous, the results of studies will become more exact and useful. Thus, a valuable future study can be a field experiment covering the relationship between personal finance and its corresponding financial behavior. Future studies could broaden the scope by incorporating additional psychological characteristics. The future study can be a comprehensive study covering participants from both developed and emerging to assess similarities and differences.

Conclusion and managerial implication

In the past decades, policymakers and educators have focused on programs to promote responsible financial behavior among households, both from developed and emerging countries. While the majority of studies are from a developed country standpoint, the present study from India adds to the literature. Although the role of financial knowledge in responsible financial management behavior was examined, the study underscores the contribution of the financial mediator as a mediator between financial knowledge and responsible financial management behavior. Furthermore, we add to the literature by exploring the role of internal locus of control in predicting responsible financial management behavior. Consistent with the theory of planned behavior, the study highlights attitude-behavior relationship and recognizes psychological aspects (Ajzen, 1991). Additionally, the role of locus of control is consistent with transtheoretical model of behavior change which suggests that internal locus of control plays a crucial role in behavioral change (Prochaska and DiClemente, 2005). The findings validate financial risk tolerance as the moderator affects in the relationship of financial knowledge, financial attitude and internal locus of control with financial management behavior. The findings have important implications for policymakers, educators and government agencies in their efforts to enhance financial management

Table 6.
OLS Regression:
Responsible Financial
management, its
components and
demographic variables

Dependent variable	Beta coefficient	<i>t</i> value	<i>p</i> value
Gender	0.070	1.211	0.226
Age	0.217	0.448	0.000
Academic background	0.035	0.571	0.568
Occupation	0.252	4.014	0.000

efficiency. Financial knowledge courses can incorporate socio-psychological aspects to enhance individual awareness of potential hurdles in converting financial knowledge to responsible financial management behavior through financial attitude.

The experience suggests that a majority of young adults have completed their formal education and concerns are raised regarding financial literacy and financial education programs. Furthermore, diverse results are available on the long-term effectiveness of financial education efforts among the young (Fernandes *et al.*, 2014; Hastings *et al.*, 2013). Past studies have investigated subjective and objective financial knowledge. To meet the requirement of young adults, support of financial education both in a formal level and at the workplace may be beneficial. The approach can be to target financial behaviors, using appropriate financial knowledge tied to financial attitude. To help convert financial knowledge into financial attitude and then to responsible financial management behavior, it is recommended to make the programs more participative. Interesting content, including digital means, can develop higher interest among young adults. Newer technological tools, such as video content, have been successful and can be used to instil responsible financial management behavior. Educators may collaborate with employers for employer-sponsored events to engage more young adults toward responsible financial management behavior. Since financial risk tolerance moderates the relationship, individual profiling based on risk levels can be helpful.

References

- Agarwalla, S.K., Barua, S.K., Jacob, J. and Varma, J.R. (2015), "Financial literacy among working young in urban India", *World Development*, Vol. 67 No. 1, pp. 101-109.
- Ajzen, I. (1991), "The theory of planned behavior", *Organizational Behavior and Human Decision Processes*, Vol. 50 No. 2, pp. 179-211.
- Alba, J.W. and Hutchinson, J.W. (2000), "Knowledge calibration: what consumers know and what they think they know", *Journal of Consumer Research*, Vol. 27 No. 2, pp. 123-156.
- Allgood, S. and Walstad, W. (2013), "Financial literacy and credit card behaviors: a cross-sectional analysis by age", *Numeracy*, Vol. 6, pp. 1-26.
- Arrow, K.J. (1971), "Aspects of the theory of risk bearing", in *Essays in the Theory of Risk-Bearing*, pp. 90-133, Chicago: Markham Publishing, (Reprinted from K.J. Arrow (1965) Aspects of the Theory of Risk-Bearing. Helsinki: Yrjö Hahnsson Foundation).
- Atkinson, A., McKay, S., Kempson, E. and Collard, S. (2006), "Levels of financial capability in the UK: results of a baseline survey", *Financial Services Authority*, UK.
- Aydin, A.E. and Akben Selcuk, E. (2019), "An investigation of financial literacy, money ethics and time preferences among college students: a structural equation model", *International Journal of Bank Marketing*, available at: <https://doi.org/10.1108/IJBM-05-2018-0120>.
- Barsky, R.B., Juster, F.T., Kimball, M.S. and Shapiro, M.D. (1997), "Preference parameters and behavioral heterogeneity: an experimental approach in the health and retirement study", *The Quarterly Journal of Economics*, Vol. 112 No. 2, pp. 537-579.
- Borden, L.M., Lee, S.-A., Serido, J. and Collins, D. (2008), "Changing college students' financial knowledge, attitudes, and behavior through seminar participation", *Journal of Family and Economic Issues*, Vol. 29 No. 1, pp. 23-40.
- Campbell, M.C. and Goldstein, R.C. (2001), "The moderating effect of perceived risk on consumers' evaluations of product incongruity: preference for the norm", *Journal of Consumer Research*, Vol. 28 No. 3, pp. 439-49.
- Carlsson Hauff, J. and Nilsson, J. (2017), "The impact of country-of-origin cues on consumer investment behavior: the moderating influence of financial brand strength and investment management style", *European Journal of Marketing*, Vol. 51 No. 2, pp. 349-366.

-
- Carpena, F., Cole, S., Shapiro, J. and Zia, B. (2011), "Unpacking the causal chain of financial literacy", World Bank Policy Research, Working Paper Series No. WPS 5798.
- Chen, H. and Volpe, R.P. (1998), "An analysis of personal financial literacy among college students", *Financial Services Review*, Vol. 7 No. 2, pp. 107-128.
- Chin Wynne, W. (1998), "Commentary: issues and Opinion in structural equation model", *MIS Quarterly*, Vol. 22 No. 1, pp. 7-16.
- Chiu, R.K. (2003), "Ethical judgment and whistleblowing intention: examining the moderating role of locus of control", *Journal of Business Ethics*, Vol. 43 Nos 1-2, pp. 65-74.
- Churchill, G. (1979), "A paradigm for developing better measures of marketing constructs", *Journal of Marketing Research*, Vol. 16, pp. 64-73.
- Cude, B.J., Swarn, C. and Jamal, T. (2019), "Financial knowledge among Iranian investors", *International Journal of Consumer Studies*, Vol. 43 No. 6, pp. 503-513.
- Deacon, R.E. and Firebaugh, F.M. (1988), *Family Resource Management: Principles and Applications*, Allyn & Bacon, Boston, MA.
- Dew, J. and Xiao, J.J. (2011), "The financial management behavior scale: development and validation", *Journal of Financial Counseling and Planning*, Vol. 22 No. 1, pp. 43-59.
- DiClemente, C.C. and Prochaska, J.O. (1982), "Self-change and therapy change of smoking behavior: a comparison of processes of change in cessation and maintenance", *Addictive Behaviors*, Vol. 7, pp. 133-142.
- Fernandes, D., Lynch, J.G. Jr and Netemeyer, R.G. (2014), "Financial literacy, financial education, and downstream financial behaviors", *Management Science*, Vol. 60 No. 8, pp. 1861-1883.
- Fitzsimmons, V.S., Hira, T.K., Bauer, J.W. and Hafstrom, J.L. (1993), "Financial management: development of scales", *Journal of Family and Economic Issues*, Vol. 14 No. 3, pp. 257-274.
- Fletschner, D. and Mesbah, D. (2011), "Gender disparity in access to information: do spouses share what they know?", *World Development*, Vol. 39 No. 8, pp. 1422-1433.
- Funfgeld, B. and Wang, M. (2009), "Attitudes and behavior in everyday finance: evidence from Switzerland", *International Journal of Bank Marketing*, Vol. 27 No. 2, pp. 108-128.
- Godwin, D.D. (1994), "Antecedents and consequences of newlyweds' cash flow management", *Financial Counseling and Planning*, Vol. 5 No. 4, pp. 161-190.
- Godwin, D.D. and Carroll, D.D. (1986), "Financial management attitudes and behaviour of husbands and wives", *Journal of Consumer Studies & Home Economics*, Vol. 10, pp. 77-96.
- Godwin, D.D. and Koonce, J.C. (1992), "Cash flow management of low-income newlyweds", *Financial Counseling and Planning*, Vol. 3 No. 1, pp. 17-42.
- Grable, J.E., Park, J.Y. and Joo, S.H. (2009), "Explaining financial management behavior for Koreans living in the United States", *Journal of Consumer Affairs*, Vol. 43 No. 1, pp. 80-107.
- Grable, J.E. (2000), "Financial risk tolerance and additional factors that affect risk taking in everyday money matters", *Journal of Business and Psychology*, Vol. 14 No. 4, pp. 625-630.
- Gron, A. and Winton, A. (2001), "Risk overhang and market behavior", *The Journal of Business*, Vol. 74 No. 4, pp. 591-612.
- Gudmunson, C.G. and Danes, S.M. (2011), "Family financial socialization: theory and critical review", *Journal of Family and Economic Issues*, Vol. 32 No. 4, pp. 644-667.
- Gurhan-Canli, Z. and Batra, R. (2004), "When corporate image affects product evaluations: the moderating role of perceived risk", *Journal of Marketing Research*, Vol. 41 No. 2, pp. 197-205.
- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E. and Tatham, R.L. (2006), *Multivariate Data Analysis* 6, Pearson-Prentice Hall, Upper Saddle River, NJ.
- Hair, J.F., Sarstedt, M., Ringle, C.M. and Mena, J.A. (2012), "An assessment of the use of partial least squares structural equation modeling in marketing research", *Journal of the Academy of Marketing Science*, Vol. 40 No. 3, pp. 414-433.

- Hallahan, T.A., Faff, R.W. and McKenzie, M.D. (2004), "An empirical investigation of personal financial risk tolerance", *Financial Services Review*, Vol. 13 No. 1, pp. 57-78.
- Hastings, J.S., Madrian, B.C. and Skimmyhorn, W.L. (2013), "Financial literacy, financial education, and economic outcomes", *Annual Review of Economics*, Vol. 5 No. 1, pp. 347-373.
- Havlena, W. and Desarbo, W.S. (1991), "On measurement of perceived consumer risk", *Decision Sciences*, Vol. 22 No. 4, pp. 927-939.
- Hilgert, M.A., Hogarth, J.M. and Beverly, S.G. (2003), "Household financial management: the connection between knowledge and behavior", *Federal Reserve Bulletin*, Vol. 89, pp. 309-322.
- Hira, T.K. and Mugenda, O. (2000), "Gender differences in financial perceptions, behaviors and satisfaction", *Journal of Financial Planning*, Vol. 13 No. 2, pp. 86-93.
- Hira, T.K., Fanslow, A.M. and Vogelsang, R. (1992), "Determinants of satisfaction with preparation for financial emergencies", *Financial Counseling and Planning*, Vol. 3, pp. 43-62.
- Hsee, C.K. and Weber Elke, U. (1999), "Cross-National differences in risk preference and lay predictions", *Journal of Behavioral Decision Making*, Vol. 12 No. 2, pp. 165-179.
- Iacobucci, D. and Duhachek, A. (2003), "Advancing alpha: measuring reliability with confidence", *Journal of Consumer Psychology*, Vol. 13 No. 4, pp. 478-487.
- Jones, J.D., Lang, W.W. and Nigro, P.J. (2005), "Agent bank behavior in bank loan syndications", *Journal of Financial Research*, Vol. 28 No. 3, pp. 385-402.
- Jorgensen, B.L., Foster, D., Jensen, J.F. and Vieira, E. (2017), "Financial attitudes and responsible spending behavior of emerging adults: does geographic location matter?", *Journal of Family and Economic Issues*, Vol. 38 No. 1, pp. 70-83.
- Judge, T.A. and Bono, J.E. (2001), "Relationship of core self-evaluations traits—self-esteem, generalized self-efficacy, locus of control, and emotional stability—with job satisfaction and job performance: a meta-analysis", *Journal of Applied Psychology*, Vol. 86 No. 1, p. 80.
- Kim, K.T., Anderson, S.G. and Seay, M.C. (2019), "Financial knowledge and short-term and long-term financial behaviors of millennials in the United States", *Journal of Family and Economic Issues*, Vol. 40 No. 2, pp. 194-208.
- Kuntze, R., Wu, C. (Ken), Wooldridge, B.R. and Whang, Y.O. (2019), "Improving financial literacy in college of business students: modernizing delivery tools", *International Journal of Bank Marketing*, Vol. 37 No. 4, pp. 976-90.
- Kyrychenko, V. and Shumb, P. (2009), "Who holds foreign stocks and bonds?, Characteristics of active investors in foreign securities", *Financial Services Review*, Vol. 18 No. 1, pp. 1-21.
- Lefcourt, H.M. (1991), "Locus of control", in Robinson, J.P., Shaver, P.R. and Wrightsman, L.S. (Eds), *Measures of Personality and Social Psychological Attitudes*, Academic Press, San Diego, CA, pp. 413-500.
- Lim, H., Heckman, S.J., Letkiewicz, J.C. and Montalto, C.P. (2014), "Financial stress, self-efficacy, and financial help-seeking behavior of college students", *Journal of Financial Counseling and Planning*, Vol. 25 No. 2, pp. 148-160.
- Lusardi, A. and Mitchell, O.S. (2007), "Financial literacy and retirement preparedness: evidence and implications for financial education", *Business Economics*, Vol. 42 No. 1, pp. 35-44.
- Lusardi, A., Mitchell, O.S. and Curto, V. (2010), "Financial literacy among the young", *Journal of Consumer Affairs*, Vol. 44 No. 2, pp. 358-380.
- Lusardi, A., Michaud, P.C. and Mitchell, O.S. (2013), "Optimal financial knowledge and wealth inequality. NBER working paper No. 18669.Slovic P (1966), risk-taking in children: age and sex differences", *Child Development*, Vol. 37, pp. 169-176.
- Lusardi, A., Michaud, P.C. and Mitchell, O.S. (2017), "Optimal financial knowledge and wealth inequality", *Journal of Political Economy*, Vol. 125 No. 2, pp. 431-77.

-
- Mas-Colell, A., Whinston, M.D. and Green, J.R. (1995), "Microeconomic Theory", Vol. 1, Oxford University Press, New York.
- Meredith, G. and Schewe, C. (1994), "The power of cohorts", *American Demographics*, Vol. 16 No. 12, pp. 22-31.
- Metcalfe, J. (1986), "Feeling of knowing in memory and problem solving", *Journal of Experimental Psychology*, Vol. 12, pp. 288-294.
- Nejad, M.G. and Javid, K. (2018), "Subjective and objective financial literacy, opinion leadership, and the use of retail banking services", *International Journal of Bank Marketing*, Vol. 36 No. 4, pp. 784-804.
- Nelson, T.O., Gerler, D. and Narens, L. (1984), "Accuracy of feeling-of-knowing judgments for predicting perceptual identification and relearning", *Journal of Experimental Psychology: General*, Vol. 113, pp. 282-300.
- Nga Joyce, K.H., Lisa, H.L.Y. and Sellappan, R.D. (2010), "A study of financial awareness among youth", *Young Consumers*, Vol. 11 No. 4, pp. 277-290.
- Norvilitis, J.M., Merwin, M.M., Osberg, T.M., Roehling, P.V., Young, P. and Kamas, M.M. (2006), "Personality factors, money attitudes, financial knowledge, and credit-card debt in college students", *Journal of Applied Social Psychology*, Vol. 36 No. 6, pp. 1395-1413.
- O'Connor, G.E. and Kabadayi, S. (2019), "Examining antecedents of health insurance literacy: the role of locus of control, cognitive style, and financial knowledge", *Journal of Consumer Affairs*, doi: [10.1111/joca.12266](https://doi.org/10.1111/joca.12266).
- Onkvisit, S. and Shaw, J. (1987), "Self-concept and image congruence: some research and managerial implications", *Journal of Consumer Marketing*, Vol. 4 No. 1, pp. 13-23.
- Park, C.W., Gardner, M.P. and Thukral, V.K. (1988), "Self-perceived knowledge: some effects on information processing for a choice task", *American Journal of Psychology*, Vol. 101 No. 3, pp. 401-424.
- Parker, A.M., Bruin, W.B., Yoong, J. and Willis, R. (2012), "Inappropriate confidence and retirement planning: four studies with a national sample", *Journal of Behavioral Decision Making*, Vol. 25 No. 4, pp. 382-389.
- Parrotta, J.L. and Johnson, P.J. (1998), "The impact of financial attitudes and knowledge on financial management and satisfaction of recently married individuals", *Journal of Financial Counseling and Planning*, Vol. 9 No. 2, p. 59.
- Perry, V.G. and Morris, M.D. (2005), "Who is in control? The role of self-perception, knowledge, and income in explaining consumer financial behavior", *Journal of Consumer Affairs*, Vol. 39 No. 2, pp. 299-313.
- Perry, V.G. (2008), "Giving credit where credit is due: the psychology of credit ratings", *The Journal of Behavioral Finance*, Vol. 9 No. 1, pp. 15-21.
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y. and Podsakoff, N.P. (2003), "Common method biases in behavioral research: a critical review of the literature and recommended remedies", *Journal of Applied Psychology*, Vol. 88 No. 5, p. 879.
- Pratt, J.W. (1964), "Risk aversion in small and large", *Econometrica*, Vol. 32, pp. 122-36.
- Prince, M. (1993), "Women, men, and money styles", *Journal of Economic Psychology*, Vol. 14, pp. 175-182.
- Prochaska, J.O. and DiClemente, C.C. (2005), "The transtheoretical approach", *Handbook of Psychotherapy Integration*, Vol. 2, pp. 147-171.
- Prochaska-Cue, K. (1993), "An exploratory study for a model of personal financial management style", *Financial Counseling and Planning*, Vol. 4, pp. 111-134.
- Reyers, M. (2019), "Financial capability and emergency savings among South Africans living above and below the poverty line", *International Journal of Consumer Studies*, Vol. 43 No. 4, pp. 335-347.
- Ringle, C.M., Wende, S. and Becker, J.M. (2015), *SmartPLS 3. Boenningstedt*, SmartPLS GmbH, available at: <http://www.smartpls.com>.

- Rotter, J.B. (1966), "Generalized expectancies for internal versus external control of reinforcement", *Psychological Monographs: General and Applied*, Vol. 80 No. 1, pp. 1-28.
- Schachter, D.L. (1983), "Feeling of knowing in episodic memory", *Journal of Experimental Psychology: Learning, Memory, and Cognition*, Vol. 9, pp. 39-54.
- Shim, S., Barber, B.L., Card, N.A., Xiao, J.J. and Serido, J. (2010), "Financial socialization of first-year college students: the roles of parents, work, and education", *Journal of Youth and Adolescence*, Vol. 39 No. 12, pp. 1457-1470.
- Slovic, P. (1986), "Informing and educating the public about risk", *Risk Analysis*, Vol. 6, pp. 403-415.
- Sobel, M.E. (1982), "Asymptotic confidence intervals for indirect effects in structural equation models", *Sociological Methodology*, Vol. 13, pp. 290-312.
- Spector, P.E. (1982), "Behavior in organizations as a function of employee's locus of control", *Psychological Bulletin*, Vol. 91 No. 3, p. 482.
- Swamy, V. (2014), "Financial inclusion, gender dimension, and economic impact on poor households", *World Development*, Vol. 56 Nos 1-15.
- Tang, T.L. (1993), "The meaning of money: extension and exploration of the money ethic scale in a sample of university students in Taiwan", *Journal of Organizational Behavior*, Vol. 14 No. 1, pp. 93-99.
- Tang, N. and Baker, A. (2016), "Self-esteem, financial knowledge and financial behavior", *Journal of Economic Psychology*, Vol. 54, pp. 164-76.
- Tang, N. and Peter, P.C. (2015), "Financial knowledge acquisition among the young: the role of financial education, financial experience, and parents' financial experience", *Financial Services Review*, Vol. 24 No. 2, p. 119.
- Titus, P.M., Fanslow, A.M. and Hira, T.K. (1989), "Net worth and financial satisfaction as a function of house-hold money managers' competencies", *Home Economics Research Journal*, Vol. 17, pp. 309-318.
- Tokar Asaad, C. (2015), "Financial literacy and financial behavior: assessing knowledge and confidence", *Financial Services Review*, Vol. 24 No. 2, pp. 101-117.
- Tsui-Yii, S. and Sheng-Chen, K. (2014), "Determinates of financial behavior: insights into consumer money attitudes and financial literacy", *Service Business*, Vol. 8 No. 2, pp. 217-238.
- Varian, H.R. (1992), *Microeconomic Analysis*, WW Norton & Company, New York, NY.
- Wallach Michael, A. and Nathan, K. (1961), "Aspects of judgment and decision making: interrelationships and changes with age", *Behavioral Science*, Vol. 6 No. 1, pp. 23-36.
- Wang, H. and Hanna Sherman, D. (1997), "Does risk tolerance decrease with age?", *Journal of Financial Counseling and Planning*, Vol. 8 No. 2, pp. 27-31.
- Warwick, J. and Mansfield, P. (2000), "Credit card consumers: college students' knowledge and attitude", *Journal of Consumer Marketing*, Vol. 17 No. 7, pp. 617-626.
- Wilhelm, M.S., Varcoe, K. and Fridrich, A.H. (1993), "Financial satisfaction and assessment of financial progress: importance of money attitudes", *Financial Counseling and Planning*, Vol. 4 No. 1, pp. 181-198.
- Worthy, S.L., Jonkman, J. and Blinn-pike, L. (2010), "Sensation-seeking, risk-taking, and problematic financial behaviors of college students", *Journal of Family and Economic Issues*, Vol. 31 No. 2, pp. 161-170.
- Xiao, J.J., Tang, C., Serido, J. and Shim, S. (2011), "Antecedents and consequences of risky credit behavior among college students: application and extension of the theory of planned behavior", *Journal of Public Policy and Marketing*, Vol. 30 No. 2, pp. 239-245.
- Xiao, J.J., Chen, C. and Chen, F. (2014), "Consumer financial capability and financial satisfaction", *Social Indicators Research*, Vol. 118 No. 1, pp. 415-432.

Yip, U.Y. (2000), "Financial risk tolerance: a state or a trait", Unpublished master's thesis, The University of New South Wales, New South Wales, (accessed 18 April 2007).

Further reading

- Atkinson, A. and Messy, F. (2012), "Measuring financial literacy: results of the OECD / international network on financial education (INFE) pilot study", OECD Working Papers on Finance, Insurance and Private Pensions, No. 15, OECD Publishing.
- Beal, D. and Delpachitra, S. (2003), "Financial literacy among Australian University students", *Economic Papers: A Journal of Applied Economics and Policy*, Vol. 22 No. 1, pp. 65-78.
- Braunsberger, K., Lucas, A.L. and Roach, D. (2005), "Evaluating the efficacy of credit card regulation", *International Journal of Bank Marketing*, Vol. 23 No. 3, pp. 237-54.
- Chen, H. and Volpe, R.P. (2002), "Gender differences in personal financial literacy among college students", *Financial Services Review*, Vol. 11 No. 3, pp. 289-307.
- Dale, L.R. and Bevil, S. (2007), "An analysis of the current status of student debt: implications for helping vulnerable students manage debt", *Academy of Leadership Journal*, Vol. 11 No. 2, pp. 121-5.
- Delavande, A., Rohwedder, S. and Willis, R. (2008), "Preparation for retirement, financial literacy and cognitive resources", Michigan Retirement Research Center Working Paper No. 2008-190.
- Fishbein, M. and Ajzen, I. (1975), *Belief, Attitude, Intention and Behavior: An Introduction to Theory and Research*, Addison-Wesley, Reading, MA.
- Grable, J.E. and Lytton, R.H. (1999), "Financial risk tolerance revisited: the development of risk assessment instrument", *Financial Services Review*, Vol. 8 No. 3, pp. 163-181.
- Greenspan, A. (2002), "Financial literacy: a tool for economic progress", *The Futurist*, Vol. 36 No. 4, pp. 37-41.
- Haliassos, M. and Bertaut, C.C. (1995), "Why do so few hold stocks?", *The Economic Journal*, Vol. 105, pp. 1110-1129.
- Heaney, J. (2007), "Generations X and Y's internet banking usage in Australia", *Journal of Financial Services Marketing*, Vol. 11 No. 3, pp. 196-210.
- Ho, H.T. and Svein, O.O. (2012), "Certainty, risk and knowledge in the satisfaction-purchase intention relationship in a new product experiment", *Asia Pacific Journal of Marketing and Logistics*, Vol. 24 No. 1, pp. 78-101.
- Hsu, J.W. (2011), "Aging and strategic learning: the impact of spousal incentives on financial literacy", Networks Financial Institute at Indiana State University, Working Paper No. 2011-WP-06.
- Huhmann, B. (2014), "Social and psychological influences on financial literacy", in *The Routledge Companion to Financial Services Marketing*, Routledge, pp. 75-91.
- Hussainey, K. and Al-Najjar, B. (2011), "Future-oriented narrative reporting: determinants and use", *Journal of Applied Accounting Research*, Vol. 12 No. 2, pp. 123-138.
- James, J.W., Leavell, H.L. and Maniam, B. (2002), "Financial planning, managers, and college students", *Managerial Finance*, Vol. 28 No. 7, pp. 35-42.
- Lim, V.K. and Teo, T.S. (1997), "Sex, money and financial hardship: an empirical study of attitudes towards money among undergraduates in Singapore", *Journal of Economic Psychology*, Vol. 18 No. 4, pp. 369-386.
- Lusardi, A. (2004), "Saving and the effectiveness of financial education", in Mitchell, O. and Utkus, W. (Eds), *Pension Design and Structure: New Lessons from Behavioral Finance 157-184*, Oxford University Press, Cambridge.
- Lusardi, A. and Mitchell, O.S. (2014), "The economic importance of financial literacy: theory and evidence", *Journal of Economic Literature*, Vol. 52 No. 1, pp. 5-44.

- Marriott, P. (2007), "An analysis of first experience students financial awareness and attitude to debt in a post-1992", UK University, *Higher Education Quarterly*, Vol. 61 No. 4, pp. 498-519.
- Mattila, A.S. and Wirtz, J. (2002), "The impact of knowledge types on the consumer search process: an investigation in the context of credence services", *International Journal of Service Industry Management*, Vol. 13 No. 3, pp. 214-30.
- McCormick, M.H. (2009), "The effectiveness of youth financial education: a review of the literature", *Journal of Financial Counseling and Planning*, Vol. 20 No. 1, pp. 70-83.
- Rotfeld, H.J. (2008), "Financial aliteracy", *The Journal of Consumer Affairs*, Vol. 42 No. 2, pp. 306-309.
- Sheppard, B.H., Hartwick, J. and Warshaw, P.R. (1988), "The theory of reasoned action: a meta-analysis of past research with recommendations for modifications and future research", *Journal of Consumer Research*, Vol. 15 No. 3, pp. 325-343.
- Sung, J. and Hanna, S. (1996), "Factors related to risk tolerance", *Financial Counseling and Planning*, Vol. 7, pp. 11-20.
- Volpe, R.P., Chen, H. and Pavlicko, J.J. (1996), "Personal investment literacy among college students A survey", *Financial Practice and Education*, Vol. 6 No. 2, pp. 86-94.
- Walstad, W., Rebeck, K. and MacDonald, R. (2010), "The effects of financial education on the financial knowledge of high school students", *Journal of Consumer Affairs*, Vol. 44 No. 2, pp. 336-357.
- Weingarten, R.M. (2009), "Four generations, one workplace: a gen X-Y staff nurse's view of team building in the emergency department", *Journal of Emergency Nursing*, Vol. 35 No. 1, pp. 27-30.
- Wells, C. (2007), "Optimism, intertemporal choice and college student debt", *Journal of Personal Finance*, Vol. 5 No. 4, pp. 44-65.
- Xiao, J.J. and Porto, N. (2017), "Financial education and financial satisfaction", *International Journal of Bank Marketing*, Vol. 35 No. 5, pp. 805-817.
- Xiao, J.J., Shim, S., Barber, B. and Lyons, A. (2008), "Academic success and well-being of college students: financial behaviors matter", Take Charge America Institute for Consumer Financial Education and Research, Tucson, AZ, available at: http://www.cefe.illinois.edu/research/reports/AcademicSuccessandWell-BeingofCollegeStudents_112007.pdf.
- Yang, S., Markoczy, L. and Qi, M. (2007), "Unrealistic optimism in consumer credit card adoption", *Journal of Education and Psychology*, Vol. 28 No. 2, pp. 170-85.

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